

Part II

Infinite Dimensions

Chapter 6

Existence, Uniqueness, and Dynamics

In this chapter we first discuss and present results concerning the existence, uniqueness, and continuity of solutions to ODEs.

6.1 Existence and Uniqueness of Solutions to ODEs

We begin this section by proving that if the vector field is locally Lipschitz continuous, then any initial value problem has a unique solution. Before, we introduce the following lemma, which follows from the observation that by the fundamental theorem of calculus a solution to an ODE can be recast as a solution to an integral equation.

Lemma 6.1.1. *Assume $f \in C(U, \mathbb{R}^n)$ where $U \subset \mathbb{R}^n$ is open, and let $x_0 \in U$. A continuous function $\varphi: [a, b] \rightarrow U$ solves the initial value problem $\dot{x} = f(x)$, $x(t_0) = x_0$ on (a, b) with $t_0 \in (a, b)$ if and only if*

$$\varphi(t) = x_0 + \int_{t_0}^t f(\varphi(s)) ds,$$

for all $t \in [a, b]$.

Observe that since f is independent of t , there is no loss of generality in assuming that $t_0 = 0$, which for the sake of simplicity of expression will be done henceforth.

Definition 6.1.2. Consider $U \subset \mathbb{R}^n$ open. A function $f: U \rightarrow \mathbb{R}^n$ is *Lipschitz continuous* if there exists a real constant $K \geq 0$ such that, for all $x_1, x_2 \in U$,

$$\|f(x_1) - f(x_2)\| \leq K\|x_1 - x_2\|.$$

The smallest K satisfying this inequality is denoted by $Lip(f)$ and is called the *Lipschitz constant of f* .

More generally, f is *locally Lipschitz continuous* if every point in U has a neighborhood such that f restricted to that neighborhood is Lipschitz continuous.

The proof of the following proposition is left to the reader.

Proposition 6.1.3. *Let $U \subset \mathbb{R}^n$ open and let $f \in C^1(U, \mathbb{R}^n)$. Then f is locally Lipschitz continuous.*

We will use the contraction mapping theorem to prove the existence and uniqueness of solutions. Thus the first step is to identify an appropriate complete metric.

Proposition 6.1.4. *Let $J = [a, b] \subset \mathbb{R}$. Consider $C^0(J, \mathbb{R})$ with the norm*

$$\|x - y\|_\infty \stackrel{\text{def}}{=} \sup_{t \in J} \|x(t) - y(t)\|.$$

Then, $C^0(J, \mathbb{R})$ is a Banach space.

Proof. We begin by remarking that $C^0(J, \mathbb{R})$ is a vector space over $\mathbb{R}$ since if $f, g \in C^0(J, \mathbb{R})$ and $a \in \mathbb{R}$, then $f + g \in C^0(J, \mathbb{R})$ and $af \in C^0(J, \mathbb{R})$.

To show that $C^0(J, \mathbb{R})$ is a complete metric space we need to show that if $\{f_n\} \subset C^0(J, \mathbb{R})$ is a Cauchy sequence, then there exists $\hat{f} \in C^0(J, \mathbb{R})$ to which the sequence converges. We provide a proof in three steps. First, we define a function $\hat{f}: [a, b] \rightarrow \mathbb{R}$. Then, we show that $\hat{f}$ is continuous. Finally we show that $\lim_{n \rightarrow \infty} \|f_n - \hat{f}\|_\infty = 0$.

Observe that for every $x \in [a, b]$

$$|f_n(x) - f_m(x)| \leq \sup_{z \in [a, b]} |f_n(z) - f_m(z)| = \|f_n - f_m\|_\infty. \quad (6.1)$$

As a consequence, the assumption that $\{f_n\}$ is Cauchy implies that $\{f_n(x)\} \subset \mathbb{R}$ is Cauchy for each $x \in [a, b]$. Since $\mathbb{R}$ is complete we can set $\hat{f}(x) := \lim_{n \rightarrow \infty} f_n(x)$.

To show that $\hat{f}$ is continuous it suffices to show that $\hat{f}$ is continuous at each $\hat{x} \in [a, b]$. To this end let $\{x_k\} \subset [a, b]$ be a sequence such that $\lim_{k \rightarrow \infty} x_k = \hat{x}$. We need to show that $\lim_{k \rightarrow \infty} \hat{f}(x_k) = \hat{f}(\hat{x})$ or equivalently given $\epsilon > 0$, there exists $K \in \mathbb{N}$ such that if $k \geq K$, then $|\hat{f}(x_k) - \hat{f}(\hat{x})| < \epsilon$. Combining (6.1) with the assumption that $\{f_n\}$ is Cauchy allows us to conclude that for m and n sufficiently large

$$|f_n(x) - f_m(x)| < \frac{\epsilon}{3}$$

independent of $x \in [a, b]$. In addition, since this is true for all n sufficiently large

$$\lim_{n \rightarrow \infty} |f_n(x) - f_m(x)| = |\hat{f}(x) - f_m(x)| < \frac{\epsilon}{3}.$$

Finally, since f_m is continuous, given $\epsilon < \frac{\epsilon}{3}$, there exists K such that if $k \geq K$, then

$$|f_m(x_k) - f_m(\hat{x})| < \frac{\epsilon}{3}.$$

Using these three inequalities we obtain that given $\epsilon > 0$, for sufficiently large k and m

$$|\hat{f}(x_k) - \hat{f}(\hat{x})| < |\hat{f}(x_k) - f_m(x_k)| + |f_m(x_k) - f_m(\hat{x})| + |f_m(\hat{x}) - \hat{f}(\hat{x})| < \epsilon,$$

which is the desired result.

We leave it to the reader to check that $\lim_{n \rightarrow \infty} \|f_n - \hat{f}\|_\infty = 0$. $\square$

Theorem 6.1.5 (Existence and Uniqueness). *If $f: U \rightarrow \mathbb{R}^n$ is a locally Lipschitz continuous function defined on an open set $U \subset \mathbb{R}^n$ and $x_0 \in U$, then there exists a solution $\varphi: (-a, a) \rightarrow U$ for some $a > 0$ to the initial value problem*

$$\dot{x} = f(x), \quad x(0) = x_0. \quad (6.2)$$

Furthermore, φ is the unique solution in the sense that if $\psi: (-b, b) \rightarrow U$ is a solution, then φ and ψ agree on their common domain.

Proof. Since f is locally Lipschitz continuous there exists $\epsilon > 0$ such that f is Lipschitz continuous on $\overline{B_\epsilon(x_0)} \subset U$. Since $\overline{B_\epsilon(x_0)}$ is compact, it follows that f is bounded on $\overline{B_\epsilon(x_0)}$. Therefore, there are constants K and M with

$$\|f(x) - f(y)\| \leq K\|x - y\| \quad \text{and} \quad \|f(x)\| \leq M$$

for all $x, y \in \overline{B_\epsilon(x_0)}$. Choose $a > 0$ such that

$$a < \min \left\{ \frac{\epsilon}{2M}, \frac{1}{K} \right\}, \quad (6.3)$$

and set $J \stackrel{\text{def}}{=} [-a, a]$.

By Lemma 6.1.1 it is sufficient to prove the existence of φ satisfying

$$\varphi(t) = x_0 + \int_0^t f(\varphi(s)) ds, \quad (6.4)$$

for all $t \in J$.

To this end set $X := \overline{B_\epsilon(x_0)} \subset C^0(J, \mathbb{R}^n)$ where in an abuse of notation we use x_0 to denote the constant function taking the value x_0 . Define $T: X \rightarrow C^0(J, \mathbb{R}^n)$ by

$$T(\alpha)(t) \stackrel{\text{def}}{=} x_0 + \int_0^t f(\alpha(s)) ds.$$

Therefore, by the Contraction Mapping Theorem to obtain a fixed point of T it is sufficient to prove (i) that $T(X) \subset X$ and (ii) that T is a contraction.

Proof of (i): Observe that for any $\alpha \in X$

$$\begin{aligned}
\|T(\alpha) - x_0\|_\infty &= \sup_{t \in J} \left\| x_0 + \int_0^t f(\alpha(s)) ds - x_0 \right\| \\
&\leq \sup_{t \in J} \left\| \int_0^t f(\alpha(s)) ds \right\| \\
&\leq \sup_{t \in J} \int_0^t \|f(\alpha(s))\| ds \\
&\leq \sup_{t \in J} \int_0^t M ds \\
&\leq \sup_{t \in J} |t| M \\
&\leq aM \\
&\leq \frac{\epsilon}{2}.
\end{aligned}$$

Thus we have shown that

$$T\left(\overline{B_\epsilon(x_0)}\right) \subset \overline{B_{\epsilon/2}(x_0)} \subset \overline{B_\epsilon(x_0)} = X \subset C^0(J, \mathbb{R}^n). \quad (6.5)$$

Proof of (ii): Observe that given $\alpha, \beta \in X$

$$\begin{aligned}
\|T(\alpha) - T(\beta)\|_\infty &= \sup_{t \in J} \|T(\alpha)(t) - T(\beta)(t)\| \\
&= \sup_{t \in J} \left\| \int_0^t f(\alpha(s)) ds - \int_0^t f(\beta(s)) ds \right\| \\
&\leq \sup_{t \in J} \int_0^t \|f(\alpha(s)) - f(\beta(s))\| ds \\
&\leq \sup_{t \in J} \int_0^t K \|\alpha(s) - \beta(s)\| ds \\
&\leq \sup_{t \in J} \int_0^t K \sup_{\tau \in J} \|\alpha(\tau) - \beta(\tau)\| ds \\
&= K \sup_{t \in J} \int_0^t \|\alpha - \beta\|_\infty ds \\
&\leq K \sup_{t \in J} |t| \|\alpha - \beta\|_\infty \\
&\leq Ka \|\alpha - \beta\|_\infty.
\end{aligned}$$

By (6.3), $Ka < 1$, and therefore $T: X \rightarrow X$ is a contraction with contraction constant Ka .

Denote by $\varphi: J \rightarrow U$ the unique fixed point of T within X and hence the unique solution to the initial value problem (6.2) that lies in X .

We conclude the proof by considering the question of uniqueness of solutions. Consider an interval $[-b, b] \subset [-a, a]$. Since $b < a$, the exact same proof implies that there is a unique solution $\psi: [-b, b] \rightarrow U$ and $\psi \in \overline{B_\epsilon(x_0)}$ where x_0 is the constant function defined on the interval $[-b, b]$. However, φ restricted to $[-b, b]$ satisfies (6.4) and therefore $\varphi(t) = \psi(t)$ for all $t \in [-b, b]$.

It remains to be shown that φ is the unique solution over all of $C^0(J, \mathbb{R}^n)$, not just X . To demonstrate this we use a proof by contradiction. Assume that there exists a distinct solution $\psi \in C^0(J, \mathbb{R}^n)$. Since φ is the unique solution in X , $\psi \notin X = \overline{B_\epsilon(x_0)}$. Therefore, there exists $t \in [-a, a]$ such that $\|\psi(t) - x_0\| = \epsilon$. Define

$$b = \inf \{ |t| \mid t \in [-a, a] \text{ and } \|\psi(t) - x_0\| = \epsilon \}.$$

Since ψ is continuous and by definition $\psi(0) = x_0$, it follows that $0 < b < a$ and $\psi([-b, b]) \subset \overline{B_\epsilon(x_0)}$. The previous remarks imply that $\varphi(t) = \psi(t)$ for all $t \in [-b, b]$. However, this is a contradiction since $\|\psi(b) - x_0\| = \epsilon$ or $\|\psi(-b) - x_0\| = \epsilon$, but by (6.5) $\|\varphi(t) - x_0\| < \epsilon/2$ for all $t \in [-b, b]$. $\square$

We begin with the following simple result that is a Corollary to Theorem 6.1.5.

Corollary 6.1.6. *Let $f: U \rightarrow \mathbb{R}^n$ be a locally Lipschitz continuous function defined on an open set $U \subset \mathbb{R}^n$. For every $x_0 \in U$ there exists a neighborhood V of x_0 and a constant $b = b(V)$, such that for every $y \in V$ there exists a solution $\varphi(\cdot, y): (-b, b) \rightarrow U$ to the initial value problem*

$$\dot{x} = f(x), \quad x(0) = y.$$

Proof. Recall that Theorem 6.1.5 guarantees the existence of a solution $\varphi: (-a, a) \rightarrow \mathbb{R}^n$ to the initial value problem $\dot{x} = f(x)$, $x(0) = x_0$. Furthermore, the proof of Theorem 6.1.5 is based on the existence of a contraction mapping $T: \overline{B_\epsilon(x_0)} \rightarrow \overline{B_\epsilon(x_0)}$ where $\overline{B_\epsilon(x_0)} \in C^0([-a, a], \mathbb{R}^n)$ and $T(\varphi) = \varphi$. Finally, to guarantee that T is a contraction it is sufficient to choose $a < \min \{ \epsilon/2M, 1/K \}$ where

$$\|f(z) - f(x)\| \leq K\|z - x\| \quad \text{and} \quad \|f(z)\| \leq M$$

for all $x, z \in \overline{B_\epsilon(x_0)} \in \mathbb{R}^n$, cf. (6.3).

Let $\delta = \epsilon/2$ and observe that if $y \in \overline{B_\delta(x_0)}$, then

$$\|f(z) - f(x)\| \leq K\|z - x\| \quad \text{and} \quad \|f(z)\| \leq M$$

for all $x, z \in \overline{B_\delta(y)}$. Thus, if

$$0 < b < \min \{ \delta/2M, 1/K \},$$

then the proof of Theorem 6.1.5 guarantees the existence of a solution $\varphi(\cdot, y): (-b, b) \rightarrow U$ to the initial value problem $\dot{x} = f(x)$, $x(0) = y$. $\square$

Proposition 6.1.7. *Let $U \subset \mathbb{R}^n$ be an open set and assume $f: U \rightarrow \mathbb{R}^n$ is a Lipschitz continuous function with $\text{Lip}(f) = K$. If $\varphi(\cdot, x_0): J_{x_0} \rightarrow U$ and $\psi(\cdot, y_0): J_{y_0} \rightarrow U$ are solutions to the initial value problem $\dot{x} = f(x)$ with $x(0) = x_0$ and $x(0) = y_0$, respectively, then*

$$\|\varphi(t, x_0) - \psi(t, y_0)\| \leq \|x_0 - y_0\| e^{K|t|} \quad (6.6)$$

for all $t \in J_{x_0} \cap J_{y_0}$.

Proof. Let $\varphi(\cdot, x_0): J_{x_0} \rightarrow U$ and $\psi(\cdot, y_0): J_{y_0} \rightarrow U$ be solutions to the initial value problems $x(0) = x_0$ and $x(0) = y_0$, respectively. By Theorem 6.1.5 the interior of the interval $J_{x_0} \cap J_{y_0}$ contains 0. Let J be any open interval such that $0 \in J \subset J_{x_0} \cap J_{y_0}$. By Lemma 6.1.1, for all $t \in J$,

$$\varphi(t, x_0) - \psi(t, y_0) = x_0 - y_0 + \int_0^t f(\varphi(s, x_0)) - f(\psi(s, y_0)) ds.$$

Let $\alpha(t) \stackrel{\text{def}}{=} \|\varphi(t, x_0) - \psi(t, y_0)\|$. Then

$$\begin{aligned} \alpha(t) &\leq \|x_0 - y_0\| + \left\| \int_0^t f(\varphi(s, x_0)) - f(\psi(s, y_0)) ds \right\| \\ &\leq \|x_0 - y_0\| + \left| \int_0^t \|f(\varphi(s, x_0)) - f(\psi(s, y_0))\| ds \right| \\ &\leq \|x_0 - y_0\| + \left| \int_0^t K \|\varphi(s, x_0) - \psi(s, y_0)\| ds \right| \\ &\leq \|x_0 - y_0\| + \left| \int_0^t K \alpha(s) ds \right| \end{aligned}$$

since $\text{Lip}(f) = K$. Applying Gronwall's Inequality with $C = \|x_0 - y_0\|$ and $\beta(t) = K$, we obtain

$$\begin{aligned} \alpha(t) &\leq \|x_0 - y_0\| \exp \left(\left| \int_0^t K ds \right| \right) \\ &\leq \|x_0 - y_0\| e^{K|t|} \\ \|\varphi(t, x_0) - \psi(t, y_0)\| &\leq \|x_0 - y_0\| e^{K|t|}, \end{aligned}$$

for all $t \in J$, where J was an arbitrary open subinterval of $J_{x_0} \cap J_{y_0}$. A limit argument then implies that the final inequality holds for all $t \in J_{x_0} \cap J_{y_0}$.

To see that solutions are locally Lipschitz continuous with respect to initial conditions, observe that given initial values x_0 and y_0 and the interval $J = (-a, a)$ as defined above

$$\begin{aligned} \|\varphi(\cdot, x_0) - \psi(\cdot, y_0)\|_{C^0(J)} &= \sup_{t \in J} \|\varphi(t, x_0) - \psi(t, y_0)\| \\ &\leq \|x_0 - y_0\| \sup_{t \in J} e^{K|t|} \\ &= \|x_0 - y_0\| e^{Ka}. \end{aligned}$$

□

The following Corollary provides a summary of the results obtained so far.

Corollary 6.1.8. *Assume $f: U \rightarrow \mathbb{R}^n$ is a locally Lipschitz continuous function defined on an open set $U \subset \mathbb{R}^n$. For every $x_0 \in U$ and $t_0 \in \mathbb{R}$ there exists a neighborhood $V_0 \subset U$ of x_0 , an interval $J_0 \subset \mathbb{R}$ containing t_0 , and a Lipschitz continuous function $\varphi: J_0 \times V_0 \rightarrow \mathbb{R}^n$ so that for each $x \in V_0$ the function $\varphi(\cdot, x)$ is the solution to the IVP $\dot{x} = f(x)$, $x(t_0) = x$ on J_0 .*

Models that arise in applications typically depend on set of parameters Λ and often are time dependent. Thus, we are interested in solutions to differential equations that appear to take a more general form. Let $J \subset \mathbb{R}$, $U \subset \mathbb{R}^n$ and $\Lambda \subset \mathbb{R}^m$ be open sets and let $f: J \times U \times \Lambda \rightarrow \mathbb{R}^n$ be a continuous function. For fixed $\lambda \in \Lambda$ a *solution* to the differential equation

$$\dot{x} = f(t, x, \lambda) \quad (6.7)$$

is a differentiable function $\varphi: J_0 \rightarrow U$ defined on an open interval $J_0 \subset J$ such that

$$\frac{d\varphi}{dt}(t) = f(t, \varphi(t), \lambda)$$

for all $t \in J_0$. For $t_0 \in J$, $x_0 \in U$ and $\lambda_0 \in \Lambda$, the *initial value problem* associated with (6.7) requires finding a solution $\varphi(t) = \varphi(t; t_0, x_0, \lambda_0)$ to $\dot{x} = f(t, x, \lambda_0)$ satisfying $\varphi(t_0) = x_0$.

The corresponding existence and uniqueness theorem is as follows.

Theorem 6.1.9. *Let $J \subset \mathbb{R}$, $U \subset \mathbb{R}^n$ and $\Lambda \subset \mathbb{R}^m$ be open sets, and assume $f: J \times U \times \Lambda \rightarrow \mathbb{R}^n$ is a locally Lipschitz continuous function. If $(t_0, x_0, \lambda_0) \in J \times U \times \Lambda$, then there exists an open neighborhood of the form $J_0 \times U_0 \times \Lambda_0$ of (t_0, x_0, λ_0) and a unique Lipschitz continuous function $\varphi: J_0 \times U_0 \times \Lambda_0 \rightarrow \mathbb{R}^n$ such that for every $(t_1, x_1, \lambda_1) \in J_0 \times U_0 \times \Lambda_0$*

$$\varphi(\cdot, t_1, x_1, \lambda_1): J_0 \rightarrow U$$

is the solution to the initial value problem

$$\dot{x} = f(t, x, \lambda_1), \quad x(t_1) = x_1. \quad (6.8)$$

Proof. The proof follows from the realization that result can be viewed as a special case of Theorem 6.1.5. Define $F: U \times J \times \Lambda \rightarrow \mathbb{R}^{n+1+m}$ by

$$F(x, s, \lambda) = (f(s, x, \lambda), 1, 0).$$

By Theorem 6.1.5, there exists a function $\varphi: J_0 \rightarrow \mathbb{R}^{n+1+m}$ which satisfies the initial value problem

$$\begin{cases} \dot{x} = f(s, x, \lambda) \\ \dot{s} = 1 \\ \dot{\lambda} = 0 \\ (x(t_0), s(t_0), \lambda(t_0)) = (x_0, s_0, \lambda_0) \in U \times J \times \Lambda. \end{cases}$$

It is left to the reader to check that the first n components of φ solve (6.8). □

The most significant restriction on the assumptions of Theorem 6.1.5 is that f is Lipschitz. As the following example indicates, existence is possible in more general settings, however uniqueness of the solution can no longer be assumed.

Example 6.1.10. Consider the initial value problem

$$\dot{x} = 3x^{\frac{2}{3}}, \quad x(0) = 0. \quad (6.9)$$

Observe that

$$\varphi(t) \equiv 0 \quad \text{and} \quad \psi(t) \stackrel{\text{def}}{=} \begin{cases} t^3 & \text{if } t \geq 0 \\ 0 & \text{if } t \leq 0 \end{cases}$$

are two (of infinitely many) distinct solutions to (6.9) on $\mathbb{R}$.

Remark 6.1.11. It is worth noting that while we have proven the existence of a solution, it is limited to a time interval of length $2a$ where $a < \min\{\frac{\epsilon}{M}, \frac{1}{K}\}$, which could be very small. Theorem 6.1.17 provides information concerning existence over longer intervals of time.

Having established existence and uniqueness we turn to the question of the maximal time interval on which a solution is defined.

Example 6.1.12. Consider the initial value problem

$$\dot{x} = x^2, \quad x(0) = x_0, \quad (6.10)$$

where we assume that $x_0 > 0$. By Theorem 6.1.5 we know that this IVP has a locally unique solution. It is straightforward to check that

$$\varphi(t, x_0) \stackrel{\text{def}}{=} \frac{x_0}{1 - tx_0}$$

is a solution and that the maximal interval in time over which φ is defined is $-\infty < t < 1/x_0$. As motivation for the next theorem it is also worth observing that

$$\lim_{t \rightarrow 1/x_0} \varphi(t, x_0) = \infty.$$

Definition 6.1.13. Let $\varphi: I \rightarrow U$ and $\psi: J \rightarrow U$ be solutions to $\dot{x} = f(x)$, where $f: U \rightarrow \mathbb{R}^n$ is a continuous function defined on an open set $U \subset \mathbb{R}^n$. We say that ψ is an *extension* of φ if $I \subset J$ and $\varphi(t) = \psi(t)$ for all $t \in I$. If we also have that I is a proper subset of J , then we say that ψ is a *proper extension* of φ .

Definition 6.1.14. A solution $\varphi: J \rightarrow U$ to $\dot{x} = f(x)$ is called a *maximal solution* if it has no proper extensions. In this case the interval J is called the *maximal interval of existence* of φ .

Theorem 6.1.15. *If $f: U \rightarrow \mathbb{R}^n$ is a locally Lipschitz continuous function defined on an open set $U \subset \mathbb{R}^n$ and $x_0 \in U$, then the initial value problem*

$$\dot{x} = f(x), \quad x(0) = x_0$$

has a unique maximal solution $\varphi: J \rightarrow U$, and the maximal interval of existence J is open.

Proof. Let J be the union of all intervals I for which there is a solution $\psi: I \rightarrow U$ for the initial value problem $\dot{x} = f(x)$, $x(0) = x_0$. Since all the intervals I contain $t_0 = 0$ we have that J is an interval. Define $\varphi: J \rightarrow U$ by $\varphi(t) \stackrel{\text{def}}{=} \psi(t)$ if $t \in I$ and $\psi: I \rightarrow U$ is a solution. Since any two solutions agree on the intersection of their domains of definition, we have that φ is well defined. The function $\varphi: J \rightarrow \mathbb{R}^n$ just defined is clearly the unique maximal solution to $\dot{x} = f(x)$, $x(0) = x_0$. It remains to show that the interval J is open. Let t_- and t_+ be the left and right end points of J , respectively. Assume that J is closed at t_+ . Then, since U is open, we can by Theorem 6.1.5 extend the solution to an interval about t_+ , which contradicts the fact that J is the maximal interval. Likewise if we assume that J is closed at t_- . Therefore J is the open interval (t_-, t_+) . $\square$

The proof of the following Lemma is left as an exercise.

Lemma 6.1.16. *Let $f: U \rightarrow \mathbb{R}^n$ be a locally Lipschitz continuous function defined on an open set $U \subset \mathbb{R}^n$. Let Q be some compact subset of U , and let $\varphi: (a, b) \rightarrow Q$ be a solution to the initial value problem $\dot{x} = f(x)$, $x(t_0) = x_0$. Define the one-sided limits*

$$\varphi(a) = x_0 + \lim_{t \rightarrow a} \int_{t_0}^t f(\varphi(s)) ds \quad \text{and} \quad \varphi(b) = x_0 + \lim_{t \rightarrow b} \int_{t_0}^t f(\varphi(s)) ds.$$

Then $\varphi: [a, b] \rightarrow Q$ is differentiable, and

$$\frac{d\varphi}{dt}(a) = f(\varphi(a)) \quad \text{and} \quad \frac{d\varphi}{dt}(b) = f(\varphi(b)).$$

Theorem 6.1.17. *Let $f: U \rightarrow \mathbb{R}^n$ be a locally Lipschitz continuous function defined on an open set $U \subset \mathbb{R}^n$. Consider the initial value problem*

$$\dot{x} = f(x), \quad x(0) = x_0$$

with maximal solution $\varphi(t, x_0)$. Let $J = (t_-, t_+)$ be the maximal interval of existence of φ .

- (i) *If $t_+ < \infty$, then given any compact set $Q \subset U$ there is a time $t_Q^+ \in (0, t_+)$ such that $\varphi(t_Q^+, x_0) \notin Q$. Similarly, if $t_- > -\infty$, then given any compact set $Q \subset U$ there is a time $t_Q^- \in (t_-, 0)$ such that $\varphi(t_Q^-, x_0) \notin Q$.*
- (ii) *If $U = \mathbb{R}^n$ and $\|f(x)\|$ is bounded, then $(t_-, t_+) = \mathbb{R}$.*

Proof. (i) Assume, by contradiction, that $Q \subset U$ is compact and $\varphi([0, t_+), x_0) \subset Q$. Since Q is compact and f is continuous, there are positive constants K_1 and K_2 such that

$$\|f(x) - f(y)\| \leq K_1\|x - y\| \quad \text{and} \quad \|f(x)\| \leq K_2$$

for all $x, y \in Q$. Indeed, a locally Lipschitz continuous function is Lipschitz continuous on any compact subset. Thus the proof of Theorem 6.1.5 implies that the solution φ satisfies $Lip(\varphi) \leq K_2$ for all $t \in [0, t_+)$.

By Lemma 6.1.16 we can extend our solution to $\varphi: [0, t_+] \rightarrow U$ such that $\frac{d\varphi}{dt}(t_+) = f(\varphi(t_+))$.

Theorem 6.1.5 guarantees the existence of $\delta > 0$ and a solution $\psi: (t_+ - \delta, t_+ + \delta)$ to the initial value problem $\dot{x} = f(x)$, $x(t_+) = x_+$. Furthermore, because $\frac{d\varphi}{dt}(t_+) = f(\varphi(t_+))$ we can construct a solution to the original initial value problem by

$$\phi(t) := \begin{cases} \varphi(t) & \text{if } t \leq t_+ \\ \psi(t) & \text{if } t \geq t_+. \end{cases}$$

This contradicts the assumption that $J = (t_-, t_+)$ is the maximal interval of existence of φ .

The argument for t_- is similar.

(ii) Letting $K \stackrel{\text{def}}{=} \sup_{x \in \mathbb{R}^n} \|f(x)\| < \infty$, note that

$$\|\varphi(t, x_0) - x_0\| = \left\| \int_0^t f(\varphi(s, x_0)) ds \right\| \leq \left| \int_0^t \|f(\varphi(s, x_0))\| ds \right| \leq K|t|. \quad (6.11)$$

This implies that $\varphi(t, x_0) \in B_R(x_0)$ for all $|t| < R/K$. Suppose that $t_+ < \infty$. Consider the compact set $Q \stackrel{\text{def}}{=} \overline{B_{R_0}(x_0)}$ with $R_0 \stackrel{\text{def}}{=} Kt_+$. From part (1), there is a time $t_Q^+ \in (0, t_+)$ such that $\varphi(t_Q^+, x_0) \notin Q$. Now, $t_Q^+ < t_+ = R_0/K$. By (6.11), we get that $\varphi(t_Q^+, x_0) \in B_{R_0}(x_0) \subset Q$. This is a contradiction, implying that $t_+ = \infty$. Similarly, one can show that $t_- = -\infty$. We conclude that $J = \mathbb{R}$. $\square$

Based on Theorem 6.1.17, from now on we will interpret φ as being naturally defined on a set of the form $\{(t, x_0) \mid x_0 \in U, t \in J_{x_0}\}$, where J_{x_0} is the maximal interval of existence. We now prove that φ is locally Lipschitz continuous with respect to the initial data.

Theorem 6.1.18. *Suppose that $f: U \rightarrow \mathbb{R}^n$ is a locally Lipschitz continuous function, and let $x_0 \in U$. Let J be a closed subinterval of the maximal interval of existence J_{x_0} . Then, there exists an open neighborhood $V_0 \subset U$ of x_0 such that $J \subset J_x$ for all $x \in V_0$. The map $\Psi: V_0 \rightarrow C^0(J)$ given by $\Psi(x)(t) = \varphi(t, x)$ is Lipschitz continuous.*

Proof. For ϵ sufficiently small the set $Q = \bigcup_{t \in J} \overline{B_\epsilon(\varphi(t, x_0))}$ is contained in U . Since Q is compact, f is Lipschitz continuous on Q with Lipschitz constant, say, K . We can apply Proposition 6.1.7 on Q . Let $t_0 = \max_{t \in J} |t|$ and let $\epsilon_0 = \epsilon e^{-Kt_0}$. Then inequality (6.6)

implies that for $x \in B_{\epsilon_0}(x_0)$ the solution $\varphi(t, x)$ cannot leave the compact set Q for $t \in J$, hence by Theorem 6.1.17(i) the interval of existence J_x contains J . Set $V_0 = B_{\epsilon_0}(x_0)$. The map $\Psi : V_0 \rightarrow C^0(J)$ given by $\Psi(x)(t) = \varphi(t, x)$ is Lipschitz continuous, since we infer from (6.6) that

$$\|\varphi(\cdot, x_1) - \varphi(\cdot, x_2)\|_{C^0(J)} \leq e^{Kt_0} \|x_1 - x_2\|, \quad \text{for all } x_1, x_2 \in V_0. \quad \square$$

Note that since $\varphi(\cdot, x_0)$ is a solution to the IVP it is differentiable in t . One can also prove differentiability with respect to the initial conditions [10, Theorem 1.261]. This question is considered in Exercise 6.3.6.

Theorem 6.1.19. *Let $J \subset \mathbb{R}$, $X \subset \mathbb{R}^n$, and $\Lambda \subset \mathbb{R}^m$ be open sets. Let $(t_0, x_0, \lambda_0) \in J \times X \times \Lambda$. If $f \in C^1(J \times X \times \Lambda, \mathbb{R}^n)$, then there are open sets $J_0 \subseteq J$, $X_0 \subseteq X$, and $\Lambda_0 \subset \Lambda$ such that $(t_0, x_0, \lambda_0) \in J_0 \times X_0 \times \Lambda_0$ and there is a unique function $\varphi \in C^1(J_0 \times X_0 \times \Lambda_0, \mathbb{R}^n)$ such that*

$$\varphi(\cdot, x, \lambda) : J_0 \rightarrow \mathbb{R}^n$$

solves the initial value problem

$$\dot{x} = f(t, x, \lambda), \quad x(t_0) = x_0.$$

6.2 Flows

However, as is discussed in Example 5.1.4 in general we cannot expect the solution set to produce a flow. In this section we show that the nonlinearity of the ODE can be modified to produce a flow while preserving the geometry of trajectories.

Definition 6.2.1. Let $f : U \rightarrow \mathbb{R}^n$ be a Lipschitz continuous function defined on an open set $U \subset \mathbb{R}^n$. Let $\varphi(\cdot, x_0) : J \rightarrow \mathbb{R}^n$ be the maximal solution to the IVP

$$\dot{x} = f(x), \quad x(0) = x_0. \quad (6.12)$$

The *orbit* through x_0 is the set $\gamma(x_0) \stackrel{\text{def}}{=} \varphi(J, x_0) \subset \mathbb{R}^n$.

To put this into the context of differential equations consider the following Corollary that arises as a special case of Theorem 6.1.5 and an application of Theorem 6.1.17(ii).

Corollary 6.2.2. *Assume $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a bounded locally Lipschitz continuous function. Define $\varphi : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $\varphi(\cdot, x_0) : \mathbb{R} \rightarrow \mathbb{R}^n$ to be the unique solution to the initial value problem $\dot{x} = f(x)$, $x(0) = x_0$. Then, φ is a locally Lipschitz flow.*

Using [10, Theorem 1.261] and extensions thereof one can prove that if $f \in C^r$, then the flow $\varphi : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is also C^r .

The assumption that the vector field f is bounded is used to guarantee that the domain of the maximal solution is all of $\mathbb{R}$ for every initial condition. Boundedness is an extremely

strong constraint. On the other hand, in dynamical systems the focus is on the geometry of orbits as opposed to their parameterization, which allows us to circumvent this constraint. To be more specific, in Definition 6.2.1 it is possible that the maximal solution is $\varphi: J \rightarrow \mathbb{R}^n$ where $J \neq \mathbb{R}$. However, if we consider the following modified IVP,

$$\dot{x} = \frac{f(x)}{1 + \|f(x)\|}, \quad x(0) = x_0, \quad (6.13)$$

then by Theorem 6.1.17(ii) the maximal solution takes the form $\psi(\cdot, x_0): \mathbb{R} \rightarrow \mathbb{R}^n$. Observe that if $\varphi(t_0, x_0) = x_1$ and $\psi(t_1, x_0) = x_1$, then

$$\frac{d\psi(t, x_0)}{dt}(t_0) = \frac{1}{1 + \|f(x_1)\|} \frac{d\varphi(t, x_0)}{dt}(t_1),$$

which means that the tangent vectors to the two solutions are the scalar multiples of each other at every point that they have in common. This implies that both differential equations produce the same orbits. The next theorem formalizes these considerations.

Theorem 6.2.3 (Reparameterization of time). *Let $U \subset \mathbb{R}^n$ be an open set. Assume $f: U \rightarrow \mathbb{R}^n$ is C^k (with $k = \infty$ allowed). Given $x_0 \in U$, let $\varphi(\cdot, x_0): J_{x_0} \rightarrow U$ be the maximal solution for the IVP (6.12). Denote by $\gamma(x_0)$ the orbit through x_0 . Then there exists a C^k function $g: U \rightarrow (0, 1]$ such that $\mathbb{R}$ is the maximal domain of existence for any solution to*

$$\dot{x} = g(x)f(x).$$

Moreover, for each $x_0 \in U$, denote by $\psi(\cdot, x_0): \mathbb{R} \rightarrow U$ the solution for the initial value problem $\dot{x} = f(x)g(x)$, $x(0) = x_0$. Then,

$$\psi(t, x_0) = \varphi(\tau(t, x_0), x_0),$$

where $\tau(\cdot, x_0): \mathbb{R} \rightarrow J_{x_0}$ satisfies $\tau(0, x_0) = 0$, and solves the differential equation

$$\dot{\tau}(t, x_0) = g(\varphi(\tau(t, x_0), x_0)) > 0.$$

Hence, $\psi(\cdot, x_0)$ is a reparameterization of $\varphi(\cdot, x_0)$ with the same oriented solution curves, yielding the same orbit $\gamma(x_0) = \varphi(J_{x_0}, x_0) = \psi(\mathbb{R}, x_0)$.

Proof. If $U = \mathbb{R}^n$, let $g(x) \stackrel{\text{def}}{=} \frac{1}{1 + \|f(x)\|^2}$, where in this proof $\|\cdot\|$ the Euclidean norm (hence $\|\cdot\|^2$ is a C^∞ function from $\mathbb{R}^n$ to $[0, \infty)$). Then $g: U \rightarrow (0, 1]$ is a C^k function, and $\|g(x)f(x)\| \leq 1$ for all $x \in U$. From Theorem 6.1.17(ii), any solution to $\dot{x} = g(x)f(x)$ is defined on $\mathbb{R}$.

In case $U \neq \mathbb{R}^n$, then consider a function $G: U \rightarrow (0, 1]$ to be a C^∞ function such that $G(x)$ approaches 0 as x goes to the boundary of U or as $\|x\|$ goes to infinity (see Theorem 19.3.3). Moreover, we scale G so that it satisfies the bound $\sup_{x \in U} \|DG(x)\| \leq 1$,

where the row vector $DG(x)$ is interpreted as a linear functional on $\mathbb{R}^n$, hence the bound on the norm implies that $|DG(x)y|$ for any $y \in \mathbb{R}^n$ and any $x \in U$. In this case, let $g(x) \stackrel{\text{def}}{=} \frac{G(x)^2}{1+\|f(x)\|^2}$, which is a C^k function, and let $F(x) \stackrel{\text{def}}{=} g(x)f(x)$. Then, $\|F(x)\| \leq |G(x)|^2 \leq 1$ for all $x \in U$. Now, consider $x_0 \in U$ and denote by $\psi(t, x_0)$ the unique solution of the IVP $\dot{x} = F(x)$, $x(0) = x_0$. From Theorem 6.1.17(i), to show that $\psi(t, x_0)$ is defined on $\mathbb{R}$, it is enough to show that $G(\psi(t, x_0))$ does not go to 0 in finite time, or equivalently that $\frac{1}{G(\psi(t, x_0))}$ does not go to infinity in finite time. Now,

$$\frac{d}{dt} \frac{1}{G(\psi(t, x_0))} = -\frac{1}{G(\psi(t, x_0))^2} DG(\psi(t, x_0))F(\psi(t, x_0))$$

and therefore,

$$\begin{aligned} \frac{1}{G(\psi(t, x_0))} - \frac{1}{G(\psi(0, x_0))} &= -\int_0^t \frac{1}{G(\psi(s, x_0))^2} DG(\psi(s, x_0))F(\psi(s, x_0)) ds \\ &= -\int_0^t DG(\psi(s, x_0)) \frac{f(\psi(s, x_0))}{1 + \|f(\psi(s, x_0))\|^2} ds. \end{aligned}$$

This implies that

$$\left| \frac{1}{G(\psi(t, x_0))} \right| \leq \left| \frac{1}{G(x_0)} \right| + \int_0^{|t|} ds = \left| \frac{1}{G(x_0)} \right| + |t|,$$

and therefore $\frac{1}{G(\psi(t, x_0))}$ does not go to infinity in finite time. Hence, $\psi(t, x_0)$ is defined for all $t \in \mathbb{R}$.

Now, let $\tau(t, x_0) \in \mathbb{R}$ be the unique solution of the IVP

$$\dot{\tau}(t, x_0) = g(\varphi(\tau(t, x_0), x_0)), \quad \tau(0, x_0) = 0.$$

Then, $\varphi(\tau(t, x_0), x_0)$ satisfies $\varphi(\tau(0, x_0), x_0) = \varphi(0, x_0) = x_0$ and

$$\begin{aligned} \frac{d}{dt} \varphi(\tau(t, x_0), x_0) &= f(\varphi(\tau(t, x_0), x_0)) \dot{\tau}(t, x_0) \\ &= f(\varphi(\tau(t, x_0), x_0)) g(\varphi(\tau(t, x_0), x_0)) \\ &= F(\varphi(\tau(t, x_0), x_0)). \end{aligned}$$

By unicity of the solutions of the IVP $\dot{x} = F(x)$, $x(0) = x_0$, we obtain that $\psi(t, x_0) = \varphi(\tau(t, x_0), x_0)$. $\square$

6.3 Exercises

Exercise 6.3.1 (Existence via Picard's method). An alternative proof for the existence of solutions is given by Picard's method. In this exercise we use Picard's method to prove the existence of a solution to the *non-autonomous* IVP

$$\dot{x} = f(x, t), \quad x(t_0) = x_0.$$

We begin with some hypotheses. Consider an open set $U \subset \mathbb{R}^n$ and an open interval $I \subset \mathbb{R}$ which contains t_0 . Denote $D \stackrel{\text{def}}{=} U \times I$. Assume that $f: D \rightarrow \mathbb{R}^n$ is continuous and that for all $t \in I$, $f(\cdot, t): U \rightarrow \mathbb{R}^n$ is a locally Lipschitz continuous function.

The goal is to show that there exists a solution $\varphi: J = (t_0 - a, t_0 + a) \rightarrow \mathbb{R}^n$ for some $a > 0$ to the integral equation

$$\varphi(t) = x_0 + \int_{t_0}^t f(\varphi(s), s) ds, \quad \text{for all } t \in J. \quad (6.14)$$

Let $\epsilon, \delta > 0$ small enough so that $D_{\epsilon, \delta} \stackrel{\text{def}}{=} \{(x, t) \mid \|x - x_0\| \leq \epsilon, |t - t_0| \leq \delta\} \subset D$ and so that

$$\|f(x, t) - f(y, t)\| \leq K\|x - y\|, \quad \text{for all } (x, t), (y, t) \in D_{\epsilon, \delta}. \quad (6.15)$$

Let

$$M_{\epsilon, \delta} \stackrel{\text{def}}{=} \max_{(x, t) \in D_{\epsilon, \delta}} \|f(x, t)\|, \quad (6.16)$$

and let $a > 0$ such that

$$a \leq \min \left\{ \delta, \frac{\epsilon}{M_{\epsilon, \delta}} \right\}. \quad (6.17)$$

This existence time a is independent of the Lipschitz constant, cf. (6.3). Define a time interval by

$$J \stackrel{\text{def}}{=} (t_0 - a, t_0 + a). \quad (6.18)$$

For any $t \in J$, define the *Picard operator* by

$$T(x)(t) = x_0 + \int_{t_0}^t f(x(s), s) ds. \quad (6.19)$$

- (i) Show that $(T(x)(t), t) \in D_{\epsilon, a}$, for every $(x, t) \in D_{\epsilon, a}$.

This allows us to define a sequence of functions $\{x^n\}_{n \geq 0}$ with $x^n: J \rightarrow \mathbb{R}^n$ by

$$x^0(t) \equiv x_0, \quad x^{n+1}(t) = x_0 + \int_{t_0}^t f(x^n(s), s) ds, \quad n \geq 0. \quad (6.20)$$

The iterations (6.20) define what is known as the *Picard iterative process*.

- (ii) Show that for every $t \in J = (t_0 - a, t_0 + a)$,

$$\|x^n(t) - x^{n-1}(t)\| \leq M_{\epsilon, \delta} K^{n-1} \frac{|t - t_0|^n}{n!}.$$

- (iii) Show that $\{x^n\}_{n \geq 0}$ is a Cauchy sequence in the space $C^0(J)$ endowed with the supremum norm.

- (iv) Show that there exists a continuous function $\varphi: J \rightarrow \mathbb{R}^n$ that is solution of the integral equation (6.14).

Exercise 6.3.2. Consider the differential equation

$$\frac{dx}{dt} = x,$$

with the initial condition $x(0) = 1$. Compute the Picard iteration for this differential equation starting with the initial function

$$x_0(t) = 1.$$

Exercise 6.3.3. Consider the differential equation

$$\begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

with the initial condition $x(0) = 1$, $y(0) = 1$. Compute the Picard iteration for this differential equation starting with the initial function

$$\begin{pmatrix} x_0(t) \\ y_0(t) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Exercise 6.3.4. Consider the differential equation

$$\frac{dx}{dt} = 2x + x^2$$

with the initial condition $x(0) = 1/2$. Compute the Picard iteration for this differential equation starting with the initial function

$$x_0(t) = 1/2.$$

Exercise 6.3.5. Prove Proposition 6.1.3.

Exercise 6.3.6 (Differentiability of the flow with respect to initial conditions).

Let $\varphi: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the flow generated by $\dot{x} = f(x)$, where $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ and bounded on $\mathbb{R}^n$. Proposition 6.1.7 guarantees that $\varphi(t, x)$ is continuous with respect to initial conditions. The following approach shows that φ is actually differentiable with respect to x .

Observe that differentiability of φ with respect to x is equivalent to the statement that for any $t \in \mathbb{R}$, $x_0 \in \mathbb{R}^n$, there exists a unique $n \times n$ matrix $D\varphi(t, x_0)$ satisfying

$$\lim_{\|h\| \rightarrow 0} \frac{\|\varphi(t, x_0 + h) - \varphi(t, x_0) - D\varphi(t, x_0)h\|}{\|h\|} = 0. \quad (6.21)$$

Let $x_0 \in \mathbb{R}^n$ and $\gamma(t) \equiv \varphi(t, x_0)$ denote the orbit segment through x_0 . Define the matrix $D\varphi(t, x_0)$ to be the solution of the *first variation equation*,

$$\frac{d}{dt}D\varphi(t, x_0) = Df(\gamma(t))D\varphi(t, x_0), \quad D\varphi(0, x_0) = Id.$$

Use the results of Exercise 4.7.13 to establish that the matrix solving this equation exists assuming only that $\phi(t, x_0) \equiv \gamma(t)$ exists and is continuous. Now show that $D\varphi(t, x_0)$ satisfies (6.21).

Exercise 6.3.7 (Flow Box Theorem). Consider the differential equation $\dot{x} = f(x)$ where $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a C^r vector field for $r \geq 1$. A point $\tilde{x} \in \mathbb{R}^n$ is called a *ordinary point* if $f(\tilde{x}) \neq 0$. Prove that there exists a C^r change of coordinates $x = g(y)$ defined on a neighborhood of $\tilde{x}$ such that

$$\begin{cases} \dot{y}_1 = 1 \\ \dot{y}_2 = 0 \\ \vdots \\ \dot{y}_n = 0 \end{cases}$$

Hint: Without loss of generality assume that $\tilde{x} = 0$ and $f(0) = (1, 0, \dots, 0)$. Let U be a sufficiently small neighborhood of the origin. Consider the map $h: U \rightarrow \mathbb{R}^n$ defined by

$$h(y_1, \dots, y_n) := \varphi(y_1, (0, y_2, \dots, y_n))$$

where $\varphi: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the flow associated with $\dot{x} = f(x)$. Make use of the inverse function theorem Theorem 19.3.2.

Exercise 6.3.8. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a *globally* Lipschitz continuous function. Consider the initial value problem

$$\dot{x} = f(x), \quad x(0) = x_0$$

with maximal solution $\varphi(t, x_0)$. Let $J = (t_-, t_+)$ be the maximal interval of existence of φ . Show that $J = \mathbb{R}$. Hint: use Gronwall's inequality (Theorem 4.4.5).

Exercise 6.3.9. Prove Lemma 6.1.16.

Exercise 6.3.10. In Chapter 3, we introduced a method to compute branches of solutions for the problem $f(x, \lambda) = 0$, where $f: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ is a C^1 map. Let (x_0, λ_0) such that $f(x_0, \lambda_0) = 0$. Assuming that $x = x(\lambda)$, write an explicit differential equation of the form $\frac{dx}{d\lambda} = F(x(\lambda))$, where F depends on some partial derivatives of f . Under which assumptions do we have existence of a unique solution to the Cauchy problem

$$\frac{dx}{d\lambda} = F(x(\lambda)), \quad x(\lambda_0) = x_0?$$

Show how to apply Euler's first order approximation method to ODEs to recover the parameter continuation as introduced in Section 3.1.1.